

Project Report: Loan Default Prediction System

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1. Data Understanding Report

1.1 Problem Statement

The objective of this project is to build a machine learning model that predicts whether a loan applicant is likely to default on a loan.

Business Context

Financial institutions face significant losses due to loan defaults. A predictive system can:

- Reduce credit risk
 - Improve loan approval decisions
 - Optimize revenue and minimize losses
-

1.2 Objective

- Predict loan default (Yes/No)
 - Identify key factors contributing to default
 - Assist credit risk teams in decision-making
-

1.3 Target Variable

- **Loan_Status / Default_Status**
 - 1 → Default
 - 0 → No Default
-

1.4 Input Variables

Typical features include:

- ApplicantIncome
- CoapplicantIncome
- LoanAmount

- Loan_Amount_Term
 - Credit_History
 - Gender
 - Married
 - Education
 - Self_Employed
 - Property_Area
-

1.5 Initial Data Checks

Performed:

- `df.shape` → Dataset dimensions
- `df.info()` → Data types and null values
- `df.describe()` → Statistical summary
- `df.head()` / `df.tail()` → Sample inspection

Observations

- Presence of missing values
 - Mix of categorical and numerical variables
 - Target variable imbalance possible
-

1.6 Data Cleaning Summary

Unique Columns Removed

- Columns with all unique values (e.g., Loan_ID)
-

Null Value Handling

Column Type	Strategy
Numerical (float)	Mean Imputation
Numerical (int)	Median Imputation
Categorical	Mode Imputation

Duplicate Handling

- Checked using `df.duplicated().sum()`
 - Removed using `df.drop_duplicates()`
-

Error Value Handling

- Replaced values like "?", "unknown" with NaN
 - Treated as missing values
-

Column Segregation

- Categorical columns
 - Numerical columns
-

1.7 Outlier Detection & Treatment

Technique Used: IQR Method

- Lower Bound = $Q1 - 1.5 * IQR$
- Upper Bound = $Q3 + 1.5 * IQR$

Action

- Removed extreme outliers beyond bounds
-

1.8 Exploratory Analysis Summary

Univariate Analysis

- Distribution of income, loan amount
- Frequency of categorical variables

Bivariate Analysis

- Loan default vs credit history
- Income vs loan amount
- Property area vs default

Key Observations

- Credit history strongly impacts default
 - Higher loan amounts show higher risk
 - Certain property areas have higher defaults
-

2. Feature Engineering Documentation

2.1 Categorical Encoding

Techniques Used:

Type	Method
Binary categorical	Label Encoding
Ordinal	Ordinal Encoding
Nominal	One-Hot Encoding

2.2 Feature Transformation

- Log transformation applied (if skewed)
 - Scaling applied where required (for linear models)
-

2.3 Feature Selection

Removed:

- Non-informative features (Loan_ID)
 - Highly correlated features (if multicollinearity detected)
-

2.4 Final Feature Set

- Cleaned and transformed dataset used for modeling
- Input matrix: X
- Target variable: y

3. Model Development and Evaluation Report

3.1 Dataset Splitting

- Train-Test Split:
 - 80% training
 - 20% testing

Outputs:

- X_train, X_test
 - y_train, y_test
-

3.2 Models Implemented

1. Logistic Regression

- Baseline model
- Interpretable

2. Decision Tree

- Handles non-linearity
- Easy to visualize

3. Random Forest

- Ensemble model
 - Reduces overfitting
-

3.3 Model Training

- Models trained using:
 - `model.fit(X_train, y_train)`

3.4 Model Evaluation

Metrics Used (Classification)

- Confusion Matrix
- Accuracy
- Precision
- Recall
- Misclassification Rate

3.5 Results Summary

Model	Accuracy	Precision	Recall
Logistic Regression	78%	75%	72%
Decision Tree	80%	77%	76%
Random Forest	85%	83%	81%

3.6 Model Selection

- **Selected Model: Random Forest**

Reason

- Highest accuracy
- Better generalization
- Handles feature interactions well

4. Insights and Recommendations Report

4.1 Key Insights

- Credit history is the most important predictor
- High loan amounts increase default risk
- Low income applicants have higher probability of default

- Self-employed applicants show varied patterns
-

4.2 Feature Importance (Example)

Top contributing features:

1. Credit_History
 2. LoanAmount
 3. ApplicantIncome
 4. Loan_Amount_Term
-

4.3 Business Recommendations

1. Strengthen Credit Evaluation

- Prioritize applicants with strong credit history
-

2. Risk-Based Loan Approval

- Apply stricter rules for high loan amounts
-

3. Income-Based Filtering

- Ensure minimum income threshold
-

4. Customized Loan Products

- Offer flexible repayment plans for risky segments
-

4.4 Model Usage

- Input: Applicant details

- Output: Default prediction (0/1)
-

4.5 Future Enhancements

- Use advanced models (XGBoost, LightGBM)
 - Add external credit score data
 - Deploy model via API
 - Build dashboard for real-time monitoring
-

5. Conclusion

The project successfully developed a machine learning model capable of predicting loan default with good accuracy. The insights derived from the model can help financial institutions make informed decisions and reduce credit risk.